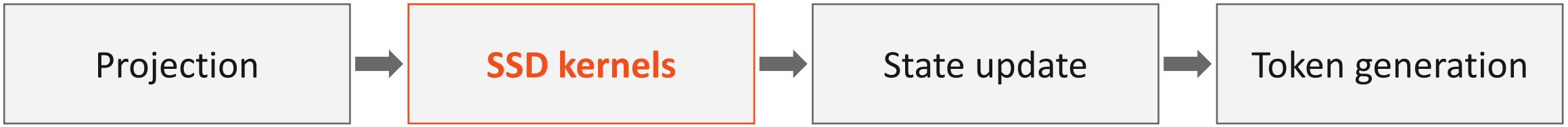
Mamba-Family State-Space Model Kernels

on a Programmable CGLA

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Mamba Is More Than One Kernel

Different phases expose different execution requirements



Long accumulation	Short reductions	Chunk dependency
Repeated FMA	Intermediate tensors	Small state updates
Large matrix work	Kernel boundaries	Transfer scheduling

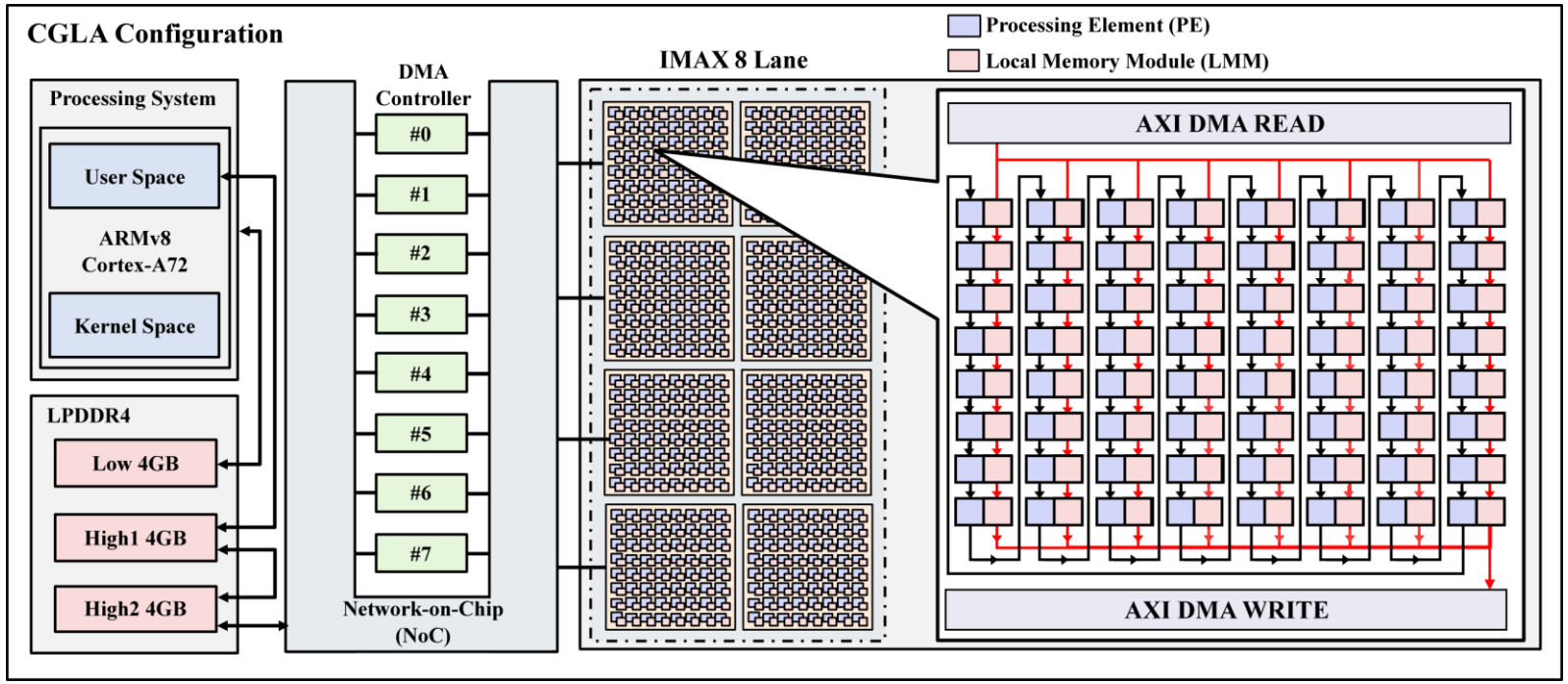
A shared MAC operation does not imply the same pipeline behavior.

Objective

Characterize Mamba-family kernels on one programmable execution stack

- Map projection, SSD Step-1, and recurrent-state kernels to IMAX.
- Separate array EXEC time from DMA-inclusive total time.
- Integrate offload into Mamba-130M token generation.
- Identify the next optimization from the measured bottleneck.

IMAX Configuration



Host / accelerator

- ARM controls DMA and kernel launch.
- PEs and LMMs form a linear pipeline.
- The compiler assigns operations to ordered rows.

CPU-Grounded Linear Array (CGLA) · a programmable ASIC architecture

FPGA prototype · AMD Versal VPK180 · 145 MHz · 64-stage pipeline

Mamba-2 Layer Computation



$$[z, x, B, C, \Delta] = u\ W_in$$

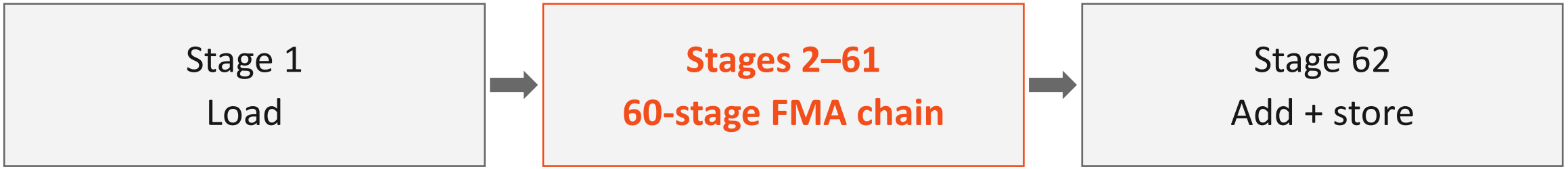
$$M = (C\ B^{T}) \odot \Lambda, \qquad Y = M\ X$$

$$y = (o \odot \sigma(z))\ W_out$$

Projection and SSD have different reduction lengths and data dependencies.

Projection Mapping

Long reductions use the existing GEMM template



Mamba-370M phase	Reduction length
in-proj	1024
out-proj	2048

Long reductions amortize pipeline fill and drain.

Stages 0 and 63 provide iteration control. Stage 63 is not an SSM recurrence loop.

SSD Step-1 Mapping

Three separate kernels



7-stage pipeline
Reduction length N

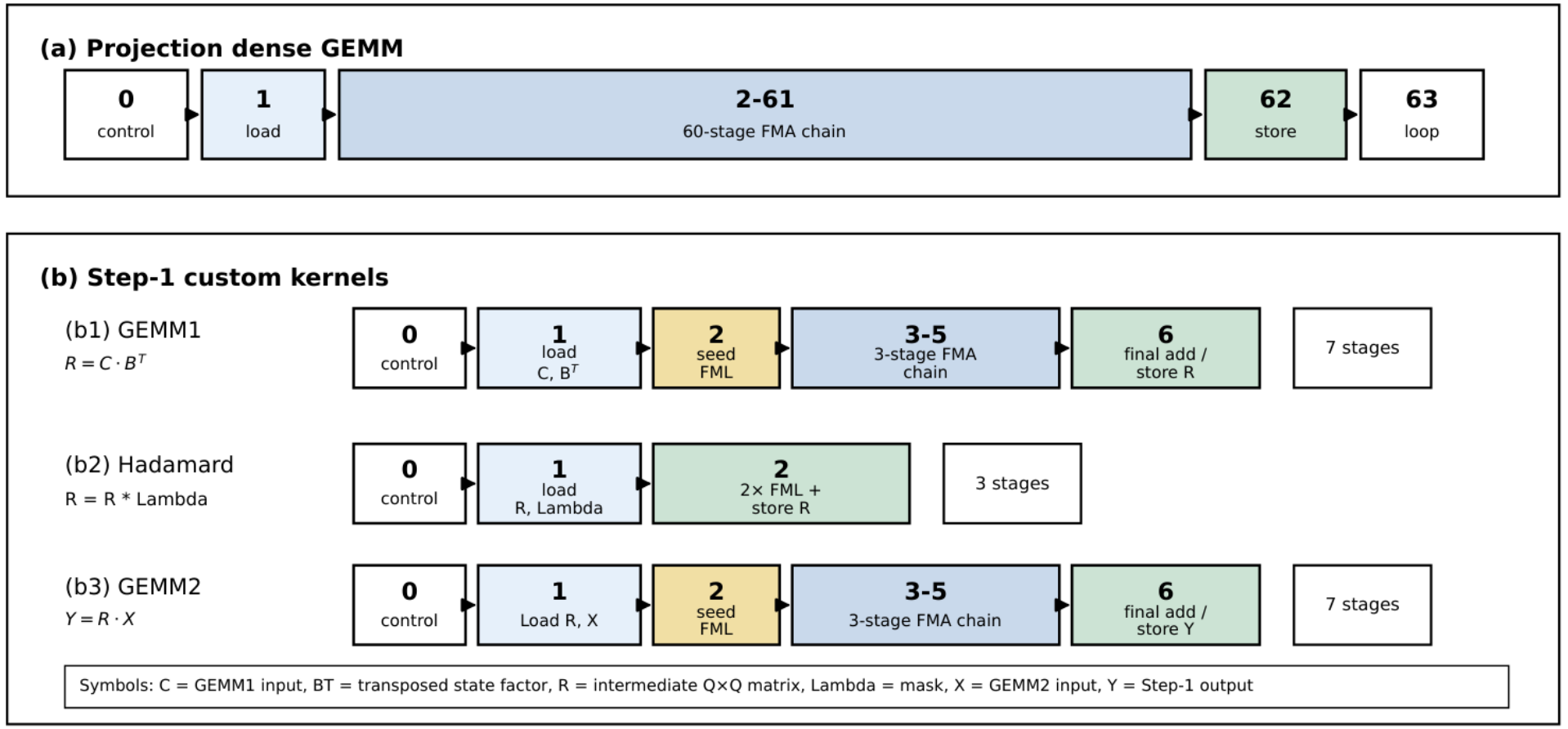
3-stage pipeline
Two FP32 lanes

7-stage pipeline
Reduction length Q

Short reductions make launch and transfer boundaries more visible.

Representative measured Step-1 configuration uses N = Q = d_head = 64.

PE Stages Used by Each Kernel



Projection uses a deep FMA chain; Step-1 uses shallow kernels.

Recurrent State Update

$$h_c = A_c\, h_(c-1) + B_c^{T}\, X_c$$



- The next chunk depends on the previous state.
- Small per-head work makes transfer scheduling important.

Step-3 offload is evaluated separately from the layer composition.

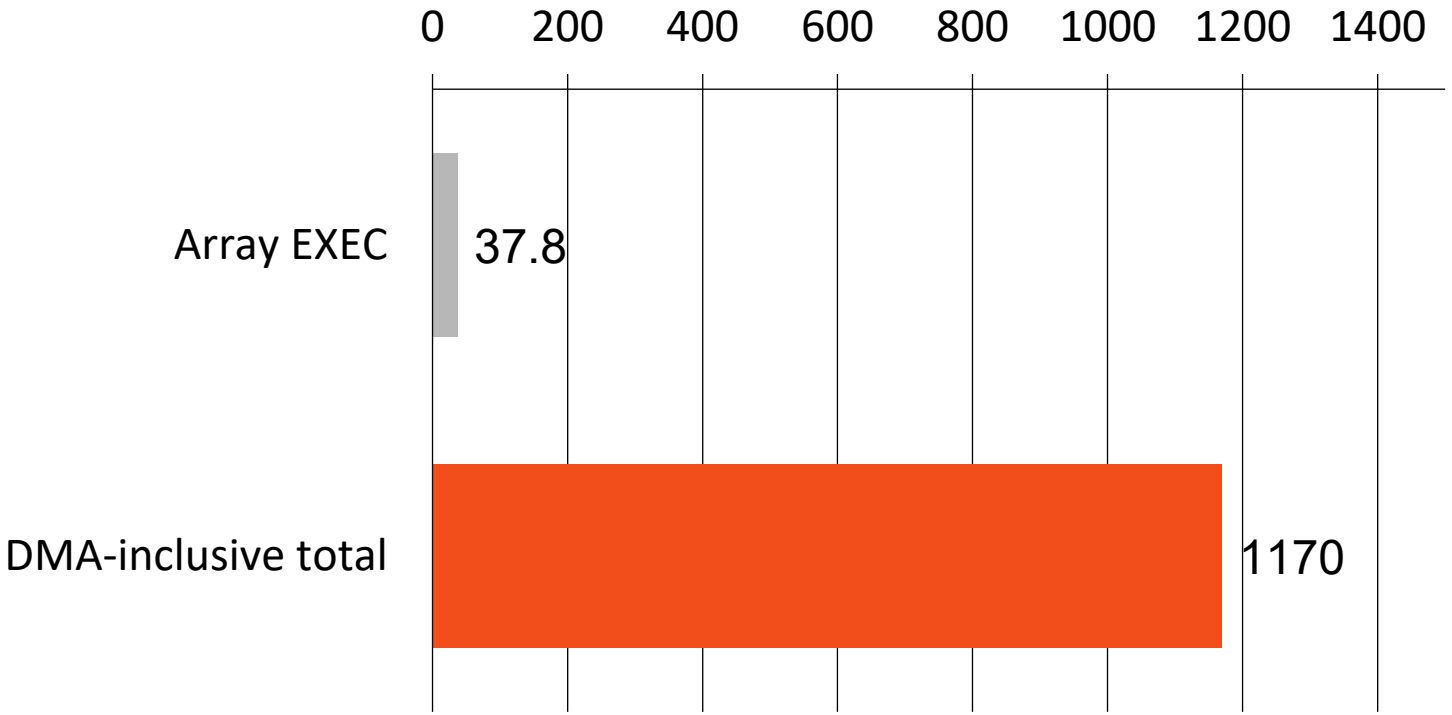
Experimental Setup

Item	Setting
Prototype	One 64-stage lane, 145 MHz, reported 512 KiB LMM
Arithmetic / reference	FP32 / co-located ARM Cortex-A72
Layer characterization	Mamba-370M; pipeline-aligned T = 120, Q = 60
Step-1 example	N = Q = d_head = 64, batch 16
Token integration	Mamba-130M, 200 generated tokens, greedy decoding
Statistics	Projection means of 3 runs; other results representative

Report EXEC, DMA-inclusive time, and token-level behavior separately.

Projection Execution and Transfer

Measured in-proj latency (ms)



37.8 ms EXEC

1.17 s total

Projection fits the long FMA chain.

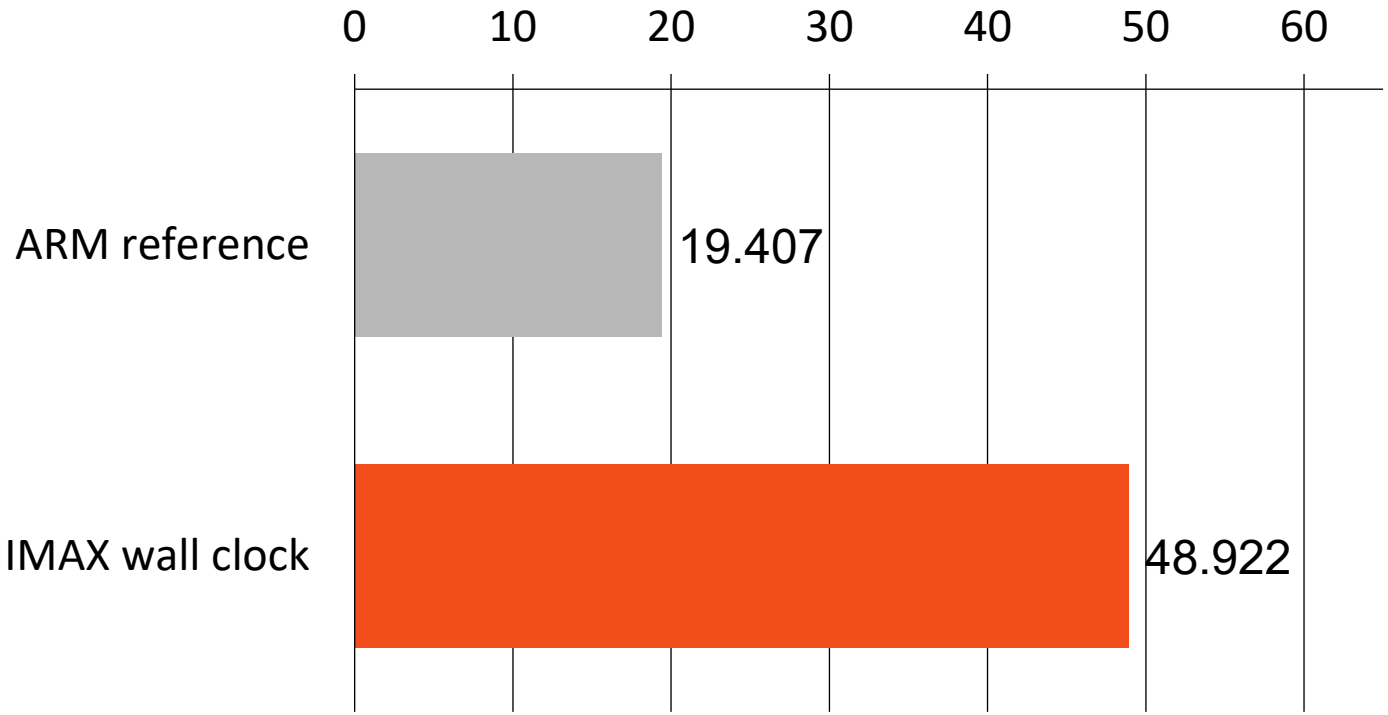
DMA dominates the observed total.

Pipeline suitability alone does not establish end-to-end acceleration.

Three archived runs; DMA-inclusive min–max spread below 0.1% for both projection kernels.

SSD Step-1 Boundary Cost

Matched Step-1 latency (ms)



Kernel	EXEC (ms)
GEMM1	7.068
Hadamard	1.178
GEMM2	7.052

Q = N = d_head = 64
Batch 16, FP32

The unfused three-kernel path is slower than the matched ARM reference.

The separate largest single-GEMM sweep gives 1.43× EXEC-only speedup, not an E2E Step-1 gain.

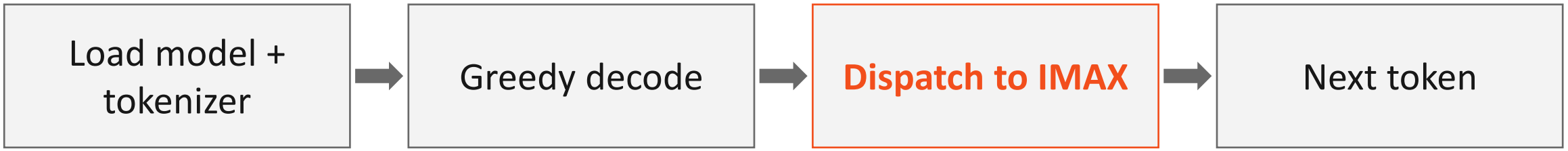
Recurrent-State Block Tuning

Workload	Default	Tuned	Ratio
Mamba-370M	29.3 ms	28.1 ms	1.04×
Mamba-1.4B	124.4 ms	107.0 ms	1.16×

Q = 64, N = 128, d_head = 64; inline DMA included.
Maximum absolute output difference from ARM is 3.46×10^{-3}.

The 1.04× change needs run-to-run variance before calling it a gain.

Token-Level Integration



Mamba-130M

200-token generation

0.345 tokens/s

ARM reference greedy sequence preserved.

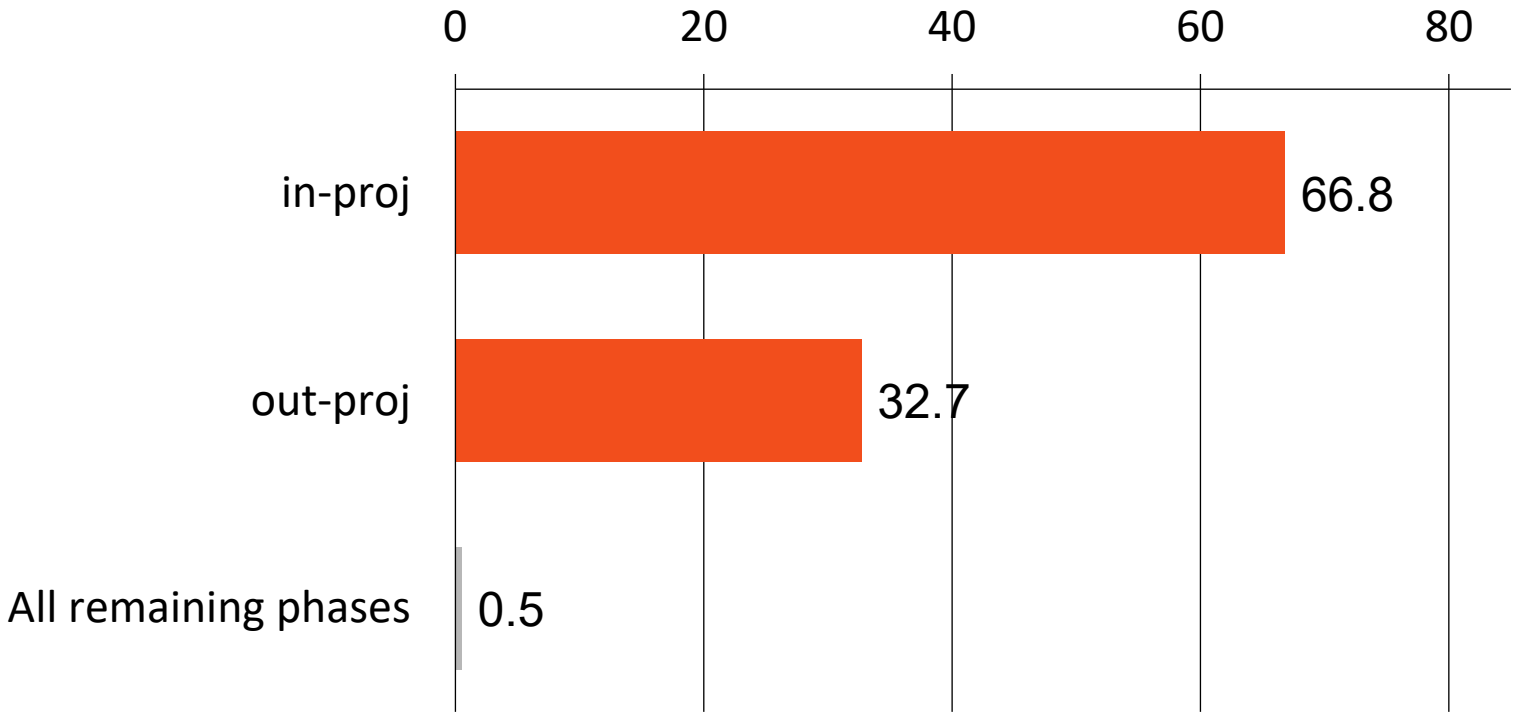
Zero reported output errors.

The run verifies token generation with the hardware-dispatch path.

No matched end-to-end CPU/GPU throughput measurement under the same program and run conditions.

What Dominates Decode?

Share of decode time (%)



Projection GEMV is the dominant decode bottleneck.

Each other phase is below 0.3% in the reported run.

Prioritize projection weights and data movement for token generation.

Remaining 0.5% is the residual of the rounded published shares, not a separately reported measurement.

Optimization Targets

Observed bottleneck	Next experiment
Projection GEMV in decode	Persistent transposed weights and wider row grouping
Step-1 kernel boundaries	Fuse Hadamard with GEMM2
FP32 transfer volume	Evaluate reduced precision under a correctness gate
Unmatched token baseline	Matched ARM / Jetson full-program measurements

Choose the optimization at the boundary that dominates the measured time.

Summary

One programmable stack, three different kernel behaviors

- Long-reduction projection matches the IMAX pipeline, but DMA dominates total time.
- Unfused SSD Step-1 loses time at short reductions and kernel boundaries.
- Mamba-130M reaches 0.345 tokens/s with the reference greedy sequence.
- Projection GEMV, rather than the state update, dominates current decode.



Thanks!

Do you have any questions?

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Backup · Nominal Layer FLOP Distribution

Phase	MFLOP at T = 512
in-proj	4598
out-proj	2147
SSD GEMM1	268
SSD GEMM2	134
Recurrent state	268
Total / projection share	7415 / 91%

FLOP share is not the same as end-to-end time share.

Mamba-370M nominal accounting, Q = 64; measured layer composition uses T = 120, Q = 60.

Backup · Measurement Scope

Boundary	What the results support
Arithmetic	FP32 only; no FP16/BF16/quantized performance claim
ASIC model	Core synthesis context only, excluding host, DDR, DMA, board
Projection	Three-run means with reported DMA-inclusive spread
Step-1 / token run	Representative measurements; no confidence interval
System acceleration	Requires a matched full-program baseline

The study characterizes kernel fit and bottlenecks on the FPGA prototype.